



Operating Instructions

Battery Energy Storage System (BESS)
mtu EnergyPack QL
2nd Generation

EPS15001/04E

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1 Preface

1.1 Preface

These Operating Instructions contain general instructions for the proper and safe operation of your Rolls-Royce Solutions Battery Energy Storage System (BESS) from Rolls-Royce Solutions.

Keep these Operating Instructions where they can be accessed by all persons entrusted with installing, operating or servicing the BESS.

Read these Operating Instructions carefully before starting any work. Follow all the safety and handling instructions to assure safe operation of the BESS.

In addition to the Operating Instructions, the following documents included in the delivery must be observed:

- Technical data sheet
- Wiring diagrams
- Foundation plan
- Fluids and Lubricants Specifications

Local accident prevention regulations and general safety regulations for the field of application of the BESS also apply.

These Operating Instructions form part of the BESS and refer to the latest design version of the product at the time of publication. They are subject to modification to reflect technical enhancements. Actual product may vary from figures shown.

For information on optional components which are not installed in this particular BESS: (This is not applicable to your system.).

2 Safety

2.1 Important requirements for the Battery Energy Storage System

General information

This product may pose a risk of injury or damage in the following cases:

- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made, nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

The residual risk of battery fires cannot be completely eliminated.

The Battery Energy Storage System (BESS) must be operated over its entire life cycle according to the conditions defined as “intended use” (→ Page 8).

The operator shall be held solely responsible for allowing non-specialist personnel access to the BESS which is otherwise strictly prohibited.

Data plate

The Battery Energy Storage System (BESS) is identified by its model designation and the serial number on the data plate. The model designation must correspond to the information provided in these Operating Instructions.

2.2 Intended use of the Battery Energy Storage System

Intended use

The Battery Energy Storage System (BESS) is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

The BESS must be operated outdoors.

The BESS must not be operated in potentially explosive atmospheres.

The BESS is intended for stationary industrial use on land.

Personnel must be trained and qualified as required by the manufacturer (→ Page 9).

The BESS must be operated within the approved ranges specified in the technical data (→ Technical data sheet).

The BESS may only be operated with approved consumables as listed in the (→ Fluids and Lubricants Specifications of the manufacturer).

The BESS must be operated in its original ex-works configuration or in a configuration which has been expressly approved in writing by the manufacturer.

The BESS must be operated in compliance with all safety regulations observing all safety and warning notes in this manual.

The BESS must be maintained at the intervals specified in the (→ Maintenance Schedule) throughout its useful life.

The BESS may only be accessed temporarily for maintenance purposes.

The BESS must be serviced and maintained in accordance with the instructions provided in this manual.

The BESS must be operated with spare parts approved by the manufacturer.

Any other use, particularly misuse, is deemed a violation of intended purpose. Any such improper use increases the risk of injury and damage when working with the BESS. The manufacturer shall not be held liable for any damage resulting from improper use.

The Fluids and Lubricants Specifications of the manufacturer are revised or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Reasonably foreseeable misuse

The BESS is not intended for mobile deployment.

The BESS must not be connected to low-voltage distribution networks for residential areas.

The BESS must not be operated on board ships.

Modifications or conversions

Unauthorized changes to the BESS breach the conditions of its intended use and compromise safety.

Changes or modifications shall only be deemed compliant with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements for working on the Battery Energy Storage System

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the Battery Energy Storage System (BESS) such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on handling and repairing the BESS based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is particularly important in the case of personnel who only occasionally perform work on or around the BESS. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the BESS must be carried out by trained, instructed and qualified personnel only:

- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work on live lines:
 - Basic requirements as per DIN VDE 0105-100 (Operation of electrical systems) or equivalent national requirements.
 - Work on live components (battery system): Qualified personnel certified to work on live lines (AuS-Pass) or equivalent national certification
 - Accessing and opening control cabinets: Trained and qualified electrician duly authorized to carry out work
- Training at the manufacturer's Training Center focusing on:
 - System overview – BESS design
 - Introduction and instruction on component parts of the BESS
 - Requirements for accessing and working on the BESS
 - Troubleshooting and remedial action

Inexperienced and untrained personnel shall be supervised by a trained and qualified electrician at all times.

The user shall define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

No access for persons with pacemakers or other active implanted medical devices (AIMDS). Personnel must not report for duty under the influence of alcohol, drugs or strong medication. Smoking strictly prohibited. No open flame.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, hearing protection, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

Wear protective clothing as per IEC 61482 Class 1 or equivalent when working on electrical components which can be de-energized.

Wear special protective clothing as per IEC 61482 Class 2 or equivalent when working on live components.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of Chemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

2.4 Safety regulations for commissioning and operating the Battery Energy Storage System

Safety regulations for commissioning

Plan, install and connect the Battery Energy Storage System (BESS) correctly and carry out acceptance in accordance with the manufacturer's specifications before putting the BESS into operation. All necessary approvals must be granted by the relevant authorities and all requirements for commissioning must be fulfilled.

Ensure that the container is integrated in the customer's lightning protection/grounding concept prior to commissioning.

Ensure that the customer has installed the following components prior to commissioning:

- If no Rolls-Royce Solutions fire extinguishing system is installed, the customer must provide for and install an alternative based on his own risk assessment.
- The flooding pipe must be connected as necessary depending on the customer's safety concept.

Ensure that the following work has been completed prior to commissioning:

- Container is seated and secured on the foundation.
- Power cables are connected and cable bushing sealed.
- Ground connections are connected and have been tested.
- Communication lines are connected and cable bushing sealed.
- Lightning protection is installed.

Whenever the BESS is subsequently put into operation ensure that:

- All maintenance and repair work has been completed.
- All protective devices are in place.
- Check battery room for condensation and dry as necessary. Never operate the BESS if condensation is visible.
- The electrical connections of all battery modules have been checked.
- All components are properly grounded. An inspection/insulation test has been conducted before putting the equipment into operation.
- The BESS is free of damage, this applies in particular to the battery modules and the cabling.

Immediately after putting the BESS into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety regulations for operation

Do not touch the BESS unless expressly instructed to do so following a written procedure.

Only access the BESS for temporary maintenance purposes.

Some parts are electrically live during operation. The following requirements must be fulfilled before putting the BESS into operation:

- The BESS may only be operated with the protective devices installed.
- The BESS may only be operated with guard covers over the live components.
- The BESS may only be operated with the doors closed.
- Keep the container free of dirt and moisture. Wipe up leaked fluids and lubricants completely. Repair leaks immediately.
- Keep weather guards clear (e.g. tarpaulins, leaves, snow).
- Never walk on the roof of the BESS during operation.
- Never leave any objects up on the roof.
- Battery room doors (personnel doors) must never be blocked or otherwise obstructed.

Observe all applicable operating instructions, safety instructions, and warning notes.

Safety regulations during operation

Users must log on to the controller via the display on the control cabinet or via a connected PC. Authorized users must log off on completion, access via display or PC is otherwise disabled (stand-by).